

Xiaoling Hu

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Current Position	• Harvard Medical School, Harvard University, USA Aug. 2023 - Present <i>Postdoctoral Research Fellow</i> - Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl
Research Interests	My research interests lie in machine learning, computer vision, with applications in health-care, multimodal AI, generative AI, VLM/LLMs, and trustworthy learning systems. In particular, I am interested in: • Topological Deep Learning • Trustworthy Machine Learning • Multimodal AI and Generative AI (GenAI) • VLM/LLM Reasoning, Efficiency and Confidence
Education	• Department of CS, Stony Brook University, USA Jan. 2018 - June 2023 <i>Doctor of Philosophy</i> • Department of EE, Tsinghua University, China Sep. 2014 - June 2017 <i>Master of Engineering</i> • Department of EE, Huazhong University of Science and Technology, China Sep. 2010 - June 2014 <i>Bachelor of Science</i>
Selected Honors and Awards	• CVPR Oral (Top 1%) 2025 • Catacosinos Fellowship 2023 • ICLR Spotlight (Top 5%) 2023 • ECCV Oral (Top 2%) 2022 • ICLR Spotlight (Top 5%) 2021 • NeurIPS Travel Award 2019
Professional Service	Area Chair • Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026 • International Conference on Learning Representations (ICLR) 2027 • The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, 2027 • International Conference on Artificial Intelligence and Statistics (AISTATS) 2026, 2027 • AAAI Conference on Artificial Intelligence 2027 • The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027

- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026

Workshop and Tutorial Organizer

- MICCAI'26 workshop on *8th Workshop on GRaphs in biomedical Image anaLysis (GRAIL)* 2026
- MICCAI'25 workshop on *Topology- and Graph-Informed Imaging Informatics (TGI3)* 2025
- MICCAI'24 workshop on *The First Workshop on Topology- and Graph-Informed Imaging Informatics (TGI3)* 2024
- MICCAI'23 tutorial on *Topology-Driven Image Analysis* 2023

Seminar Organizer

- Co-organizer, CML Seminar Series, Athinoula A. Martinos Center for Biomedical Imaging

Selected Publications (* indicates equal contribution, __ denotes students (co-)mentored by me, ‡ denotes (co)-senior supervision)

- [1] **Topology-Aware Uncertainty for Efficient Vessel Annotation**
Saumya Gupta, Sumedh Ghavat, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), under major revision, 2026
- [2] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
Conference on Neural Information Processing Systems (NeurIPS), 2026
- [3] **RankByGene: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen
IEEE Transactions on Medical Imaging (TMI), 2026
- [4] **Multi-Channel Uncertainty-Weighted Score Matching for Conditional Diffusion in Medical UDA**
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen
European Conference on Computer Vision (ECCV), 2026
- [5] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [6] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**
Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen
Conference on Neural Information Processing Systems (NeurIPS), 2025

Short version selected as an **Oral Presentation** at the Imageomics Workshop at NeurIPS 2025

- [7] **Learn2Synth: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**
Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl[‡], Yaël Balbastre[‡]
IEEE/CVF International Conference on Computer Vision (ICCV), 2025
- [8] **TopoCellGen: Generating Histopathology Cell Topology with a Diffusion Model**
Meilong Xu, Saumya Gupta, Xiaoling Hu, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
(**Oral Presentation, acceptance rate < 1%**)
Short version also presented at CVPR 2025 Workshop GMCV
- [9] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**
Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti[‡], Juan Eugenio Iglesias[‡]
International Conference on Learning Representations (ICLR), 2025
- [10] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**
Meilong Xu, Xiaoling Hu, Saumya Gupta, Shahira Abousamra, Chao Chen
European Conference on Computer Vision (ECCV), 2024
- [11] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**
Peirong Liu, Oula Puonti, Xiaoling Hu, Daniel C. Alexander, Juan Eugenio Iglesias
European Conference on Computer Vision (ECCV), 2024
- [12] **Anomaly-Guided Weakly Supervised Lesion Segmentation on Retinal OCT Images**
Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, Xiaoling Hu, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai
Medical Image Analysis (MedIA), 2024
- [13] **Topology-Aware Uncertainty for Image Segmentation**
Saumya Gupta, Yikai Zhang, Xiaoling Hu, Prateek Prasanna, Chao Chen
Conference on Neural Information Processing Systems (NeurIPS), 2023
- [14] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**
Chen Li, Xiaoling Hu, Shahira Abousamra, Chao Chen
IEEE/CVF International Conference on Computer Vision (ICCV), 2023
- [15] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**
Aishik Konwer, Xiaoling Hu, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna
IEEE/CVF International Conference on Computer Vision (ICCV), 2023
- [16] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**
Xiaoling Hu, Dimitris Samaras, Chao Chen
International Conference on Learning Representations (ICLR), 2023 (**Spotlight Presentation, notable-top-25% of accepted papers**)

Short version selected as an **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2023

- [17] **Confidence Estimation Using Unlabeled Data**
Chen Li, **Xiaoling Hu**, Chao Chen
International Conference on Learning Representations (ICLR), 2023
- [18] **Structure-Aware Image Segmentation with Homotopy Warping**
Xiaoling Hu
Conference on Neural Information Processing Systems (NeurIPS), 2022
- [19] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**
Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tsapopoulos, Prateek Prasanna, Chao Chen
European Conference on Computer Vision (ECCV), 2022 (**Oral Presentation, acceptance rate 2.7%**)
- [20] **Trigger Hunting with a Topological Prior for Trojan Detection**
Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen
International Conference on Learning Representations (ICLR), 2022
- [21] **Topology-Aware Segmentation Using Discrete Morse Theory**
Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen
International Conference on Learning Representations (ICLR), 2021 (**Spotlight Presentation, acceptance rate 5.6%**)
- [22] **Topology-Preserving Deep Image Segmentation**
Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen
Conference on Neural Information Processing Systems (NeurIPS), 2019

Preprints

- [1] **ORCE: Order-Aware Alignment of Verbalized Confidence in Large Language Models**
Chen Li, **Xiaoling Hu**, Songzhu Zheng, Jiawei Zhou, Chao Chen
Tech Report, 2026
- [2] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026
- [3] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025
- [4] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen
Tech Report, 2025

- [5] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**
 Meilong Xu, Di Fu, Jiaying Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao
Tech Report, 2025
- [6] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24, ECCV'26**), Ph.D. student at Department of BMI, Stony Brook University
 Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
 Since Summer 2023
- Wentao Huang (**MICCAI'24, CVPR'26, TMI'26**), Ph.D. student at Department of CS, Stony Brook University
 Since Summer 2023
- Qingqiao Hu, Ph.D. student at Department of CS, Stony Brook University
 Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
 Since Fall 2025
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
 Summer 2023 – Summer 2024
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
 Fall 2021 – Summer 2023
- Jiaqi Yang (**ISBI'21, MICCAI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
 Spring 2020 – Summer 2023
- Luca Drole, Master's student in BME, ETH Zurich
 Spring – Fall 2025
- John Xie, high school student → University of Michigan
 Summer 2021

Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- Kahlert School of Computing & SCI Institute, University of Utah
 Mar. 2026
- School of Computing, University of Georgia
 Mar. 2026
- Department of EECS, University of Tennessee, Knoxville
 Feb. 2026
- Department of CS, Missouri S&T
 Jan. 2026
- Department of CSE, University of North Texas
 Nov. 2025
- HIT Webinar - *Healthcare, Intelligence, Technology*
 Aug. 2025

Learn2Synth: A Learnable Data Synthesis Strategy for Image Segmentation

- Nobrainer Seminar, Massachusetts Institute of Technology
 June 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver Oct. 2023
- Geometry and Topology meet Data Analysis and Machine Learning Aug. 2021

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University Apr. 2023
- Department of BMI, Ohio State University Mar. 2023
- Department of CS, Rochester Institute of Technology Feb. 2023
- Department of ECE, University of California, Riverside Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute Oct. 2022

Topology-aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) May 2021

Industry Experiences

- **Allen Institute, USA** May 2022 - Aug. 2022
Research Intern
Mentor: *Dr. Matheus Viana*
Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** May 2021 - Aug. 2021
Research Intern
Mentor: *Dr. Shanhui Sun*
Topic: Deep Shape Model Based Network
- **Tencent Youtu Lab, China** Jun. 2017 - Jan. 2018
Research Intern
Mentor: *Dr. Yuwing Tai*
Topic: Clothes Detection, Attribute Prediction